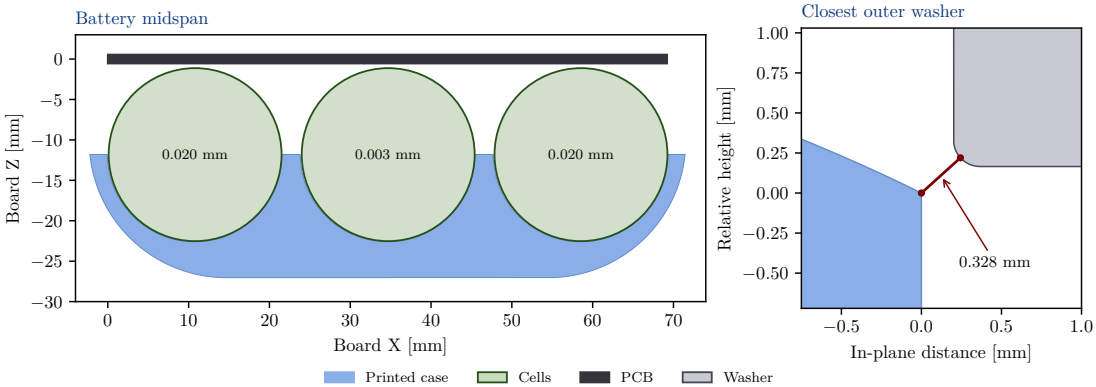
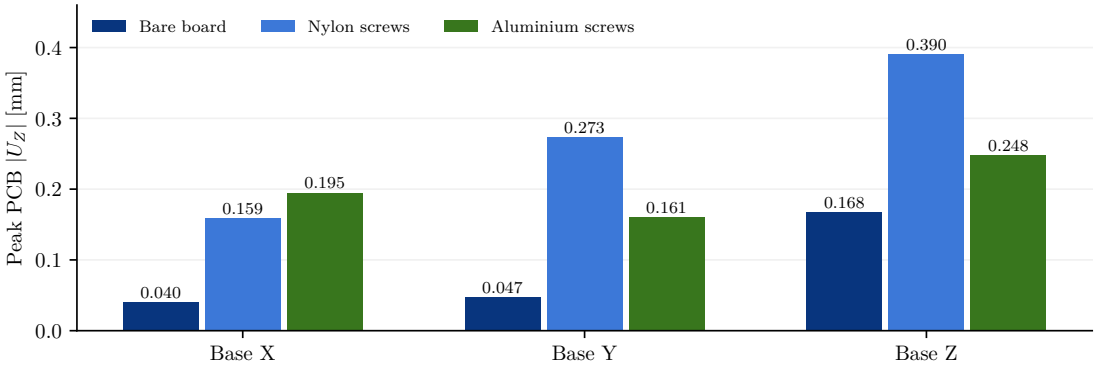
The three-pocket Balius case fits at 1.25 mm stand-off

Balius native case at 1.25 mm tab-to-board spacing



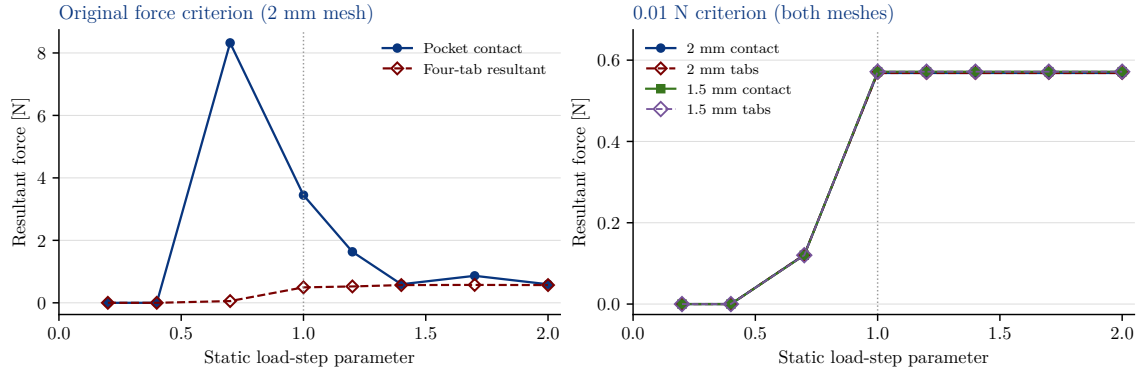
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. In the local pull model the left cell first reaches its pocket at about **1.14 N toward the middle cell** (0.020 mm) or **6.7 N away** (0.165 mm). The forces come from the bare strips up to contact.

The snug case increases board travel in the seeded clips



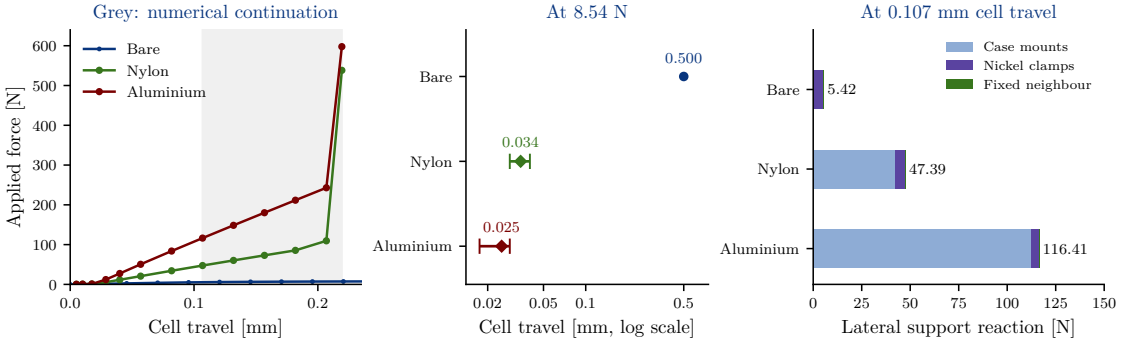
During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

Balanced nylon pocket force agrees across two meshes



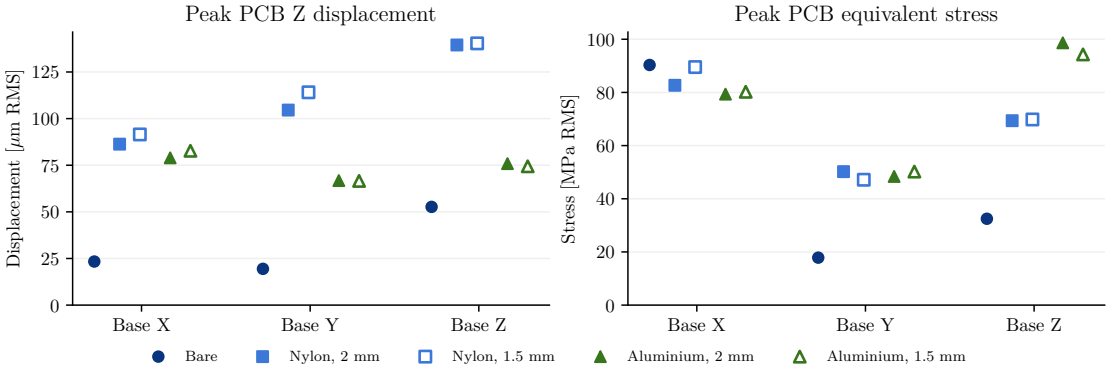
With the 0.01 N force criterion, the 2 and 1.5 mm nylon meshes give **0.5684 and 0.5716 N** at lock, a **0.55% difference**. All saved mounting-force balances pass. Local peak pressure still changes by **18.4%**. Across the tested meshes, peak PCB deflection falls about **39% with nylon** and **51% with aluminium** relative to bare.

Stiffer screws raise local restraint and contact pressure



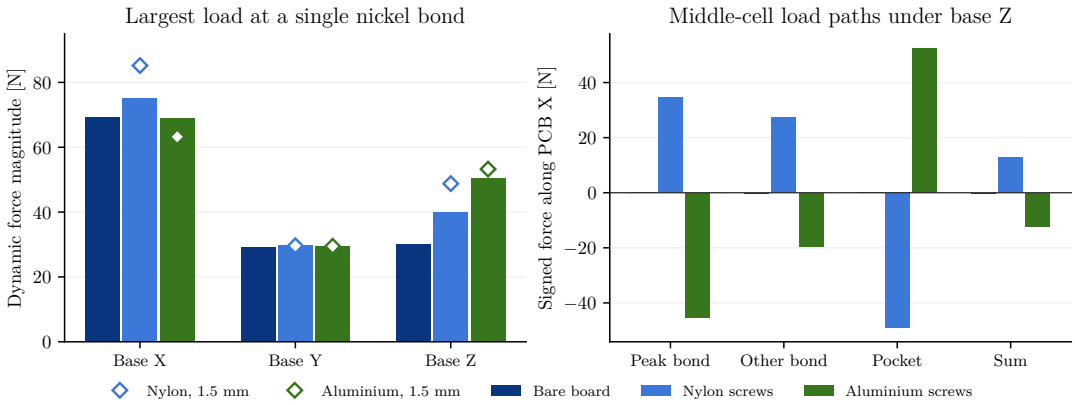
At about **0.107 mm**, local reaction is **47 N with nylon**, **116 N with aluminium** and **5.4 N bare**. Aluminium's sharp PLA corner crosses the coupon reference in the next saved interval; the crossing is not mesh-converged. Grey portions are numerical continuation. Smaller-increment controls changed matched forces by **less than 1%** for both screw materials; all 42 states per case passed balance checks. These local fixtures do not establish whole-board failure loads.

Aluminium cuts board travel relative to nylon but raises base-Z stress



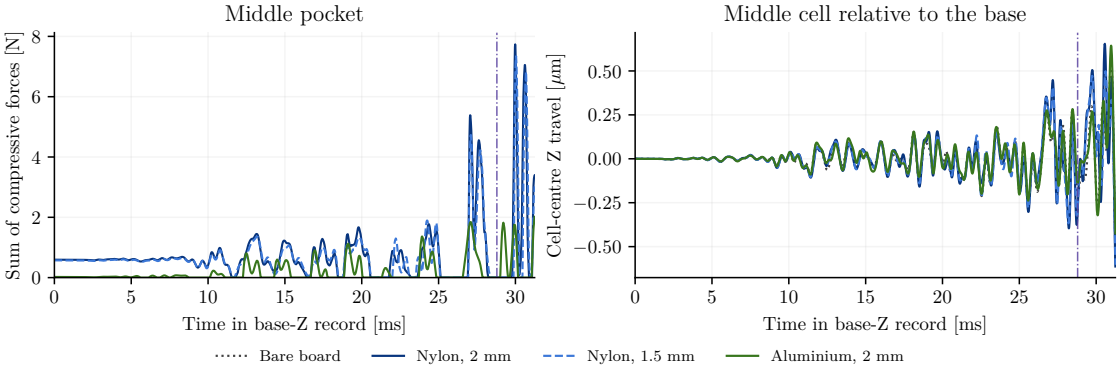
Under base Z, peak PCB stress was **32.5 MPa RMS bare**. Refining the case gave **69.8 MPa with nylon** and **94.3 with aluminium**. Filled markers are native spectra. Open markers integrate the finer native modes independently. All retain the installed contact state.

Board-normal vibration loads the middle cell's nickel bond



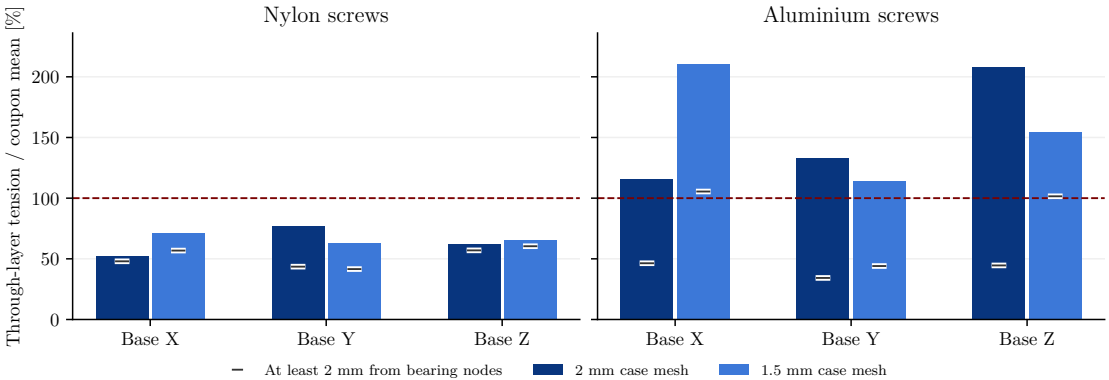
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon** and **50.5 N with aluminium**. The finer meshes gave **48.8 and 53.3 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories. Diamonds integrate the finer native modes independently.

The middle pocket unloads and carries force again



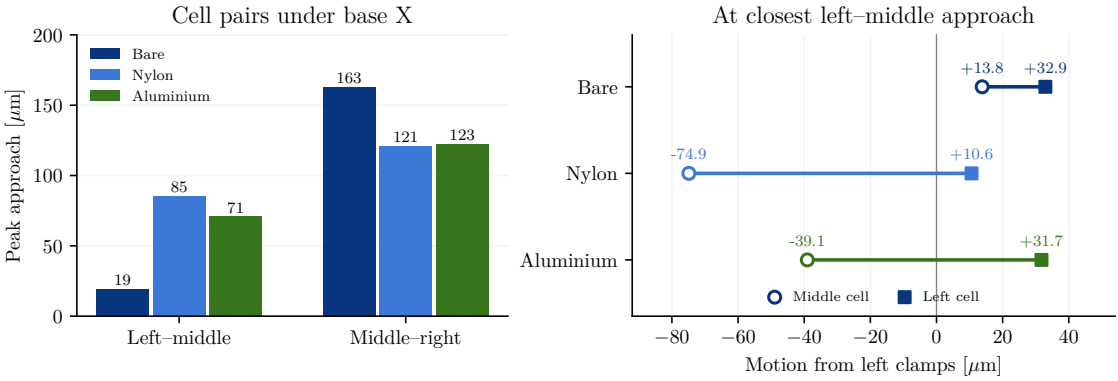
Under base Z, the original pocket points first all unloaded at **5.61 ms with aluminium** and **11.54 ms with nylon** (11.64 ms on finer nylon). At 28.79 ms the middle cell cleared the nominal meshes by **3.90–4.01 μm with nylon** and **1.73–1.84 μm with aluminium**. The tighter native cylinder check passed force balance but gave **0.044 N at lock versus 0.022 N nominal**. Local contact remains formulation-sensitive.

PLA demand exceeds its coupon reference with aluminium screws



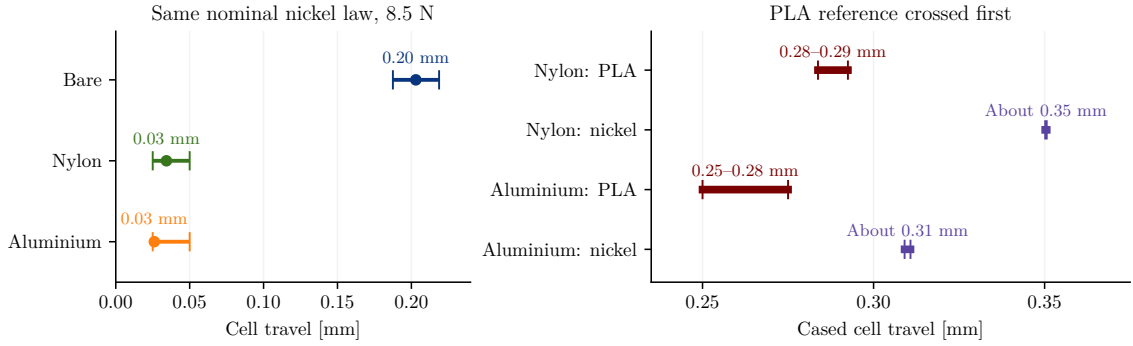
Refinement shifted the largest through-layer demand to **71.5% of the coupon mean with nylon** and **210.8% with aluminium**, both under base X. Horizontal marks exclude the 2 mm bearing band. Fine screw peaks were **37.4 and 96.2 MPa**; two integration methods agreed within 0.302%. These elastic histories retain the installed contact state.

Under base X, the left and middle cells move closer together



The left cell's peak travel relative to its clamps fell by **29% with nylon** and **12% with aluminium**. Rotation accounted for **82%** of that travel in all three assemblies. With the middle cell free to move, peak approach rose from **19 μm bare** to **85 and 71 μm**. Contact stays fixed at bolt lock in these native histories.

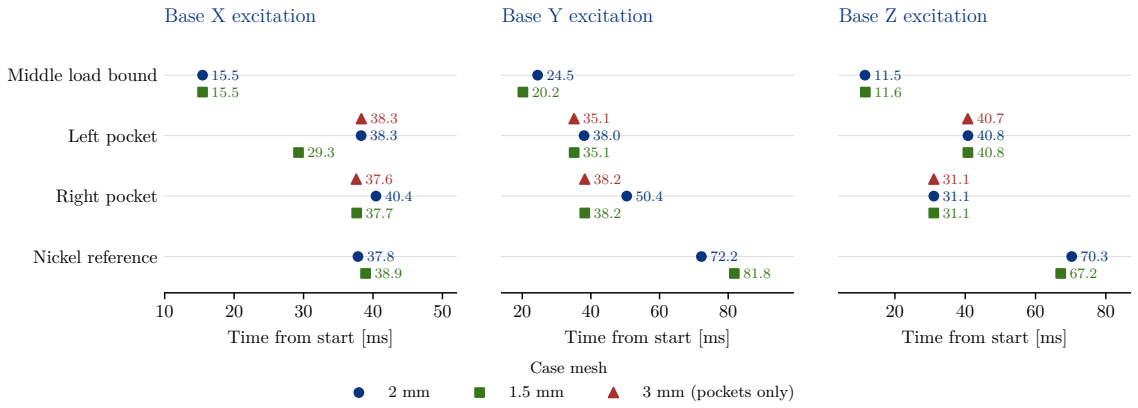
PLA crosses its reference before reported nickel damage



PLA: layer stress (nylon), layer strain (aluminium); nickel: section $D > 10^{-4}$.

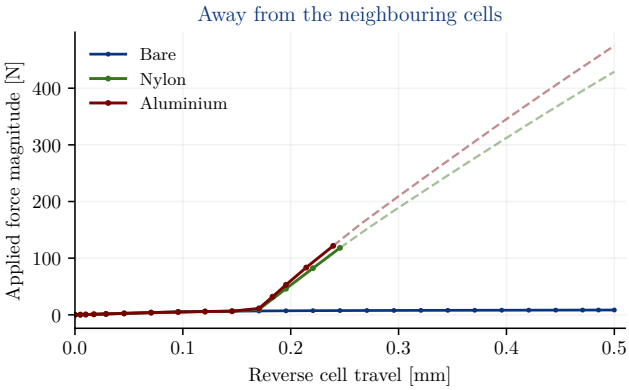
At **8.5 N**, travel is about **0.03 mm with either screw material versus 0.20 mm bare**; the cased brackets overlap. **PLA crosses its reference before nickel damage in both cases.** All 28 nylon and 23 aluminium converged states pass balance checks. Nylon stopped on nonconvergence; aluminium on excessive shell-thickness change. A tearing load remains unresolved.

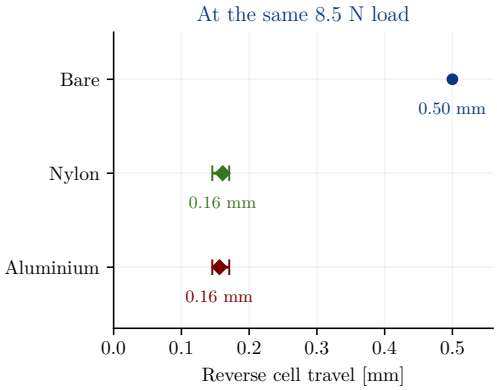
Outer-cell contact timing is still sensitive to the case mesh



Under base X, the 3 mm mesh puts the **right pocket first**; the 2 and 1.5 mm meshes put the **left first**. Base-Y timing is nonmonotonic with refinement. All shown trajectories retain the installed contact state. Both whole-shell controls stopped at native mesh-shape checks. Their contact and transient dependents remain blocked.

Both snug mounts limit reverse sideways travel





At **8.5 N**, both mounts give **0.15–0.17 mm** travel, versus **0.50 mm bare**. At the same **0.196 mm** travel, reaction is **46 N nylon**, **53 N aluminium** and **7.1 N bare**. Both native paths passed their numerical checks. The first PLA reference brackets overlap near **0.24–0.27 mm**; dashes are numerical continuation, not failure predictions.